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$$\begin{aligned}\int_0^1 \frac{1}{x^2 + x + 1} dx &= \int_0^1 \frac{1}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} d\left(x + \frac{1}{2}\right) \\&= \left[\frac{2}{\sqrt{3}} \operatorname{Bgtan} \left(\frac{2x+1}{\sqrt{3}} \right) \right]_0^1 \\&= \frac{2\pi}{3\sqrt{3}} - \frac{2\pi}{6\sqrt{3}} = \frac{\pi}{3\sqrt{3}}\end{aligned}$$

References